

Bölüm 12

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr}\right) R(r) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)\hbar^2}{2\mu r^2} + \frac{ze^2}{r} - E \right) R(r) = 0$$

$$\left(\frac{8\mu E}{\hbar^2}\right)^{1/2} \frac{d}{dp} = \frac{d}{dr}$$

$$\frac{8\mu E}{\hbar^2} \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \frac{2\mu}{\hbar^2} \left(-\frac{l(l+1)\hbar^2}{2\mu p^2} \left(\frac{8\mu E}{\hbar^2} \right) + \frac{ze^2}{p} \left(\frac{8\mu E}{\hbar^2} \right)^{1/2} - E \right) R(p) = 0$$

$$\left(\frac{8\mu E}{\hbar^2} \right) \left\{ \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \left(-\frac{l(l+1)}{p^2} + \frac{2\mu}{\hbar^2} \frac{ze^2}{p} \left(\frac{\hbar^2}{8\mu E} \right)^{1/2} - \frac{2\mu}{\hbar^2} E \left(\frac{\hbar^2}{8\mu E} \right) \right\} R(p) = 0$$

$$\left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) - \frac{l(l+1)}{p^2} R(p) + \frac{ze^2}{p^2} \left(\frac{\mu}{2E} \right) R(p) - \frac{1}{4} R(p) = 0$$

$$\frac{1}{p}$$

$$\left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \left(-\frac{l(l+1)}{p^2} + \frac{1}{p} - \frac{1}{4} \right) R(p) = 0 \stackrel{N}{=} U^2 - \frac{1}{4} = 0 \text{ Asympt. Dargması}$$

$$U^2 - \frac{1}{4} = 0 \Rightarrow U = \pm \frac{1}{2} \Rightarrow e^{\pm p/2} \quad \lim_{p \rightarrow \infty} = 0 \quad e^{+p/2} \times e^{-p/2} \quad (U - \frac{1}{2}) R(p) = 0 \Rightarrow R(p) = G(p) e^{-p/2}$$

$$R(p) = G(p) e^{-p/2} \Rightarrow R'(p) = G'(p) e^{-p/2} - \frac{1}{2} G(p) e^{-p/2}$$

$$R''(p) = G''(p) e^{-p/2} - G'(p) e^{-p/2} + \frac{1}{4} G(p) e^{-p/2}$$

$$R = G \quad R' = G' - \frac{1}{2} G \quad R'' = G'' - G' + \frac{1}{4} G \quad \left(\frac{d^2}{dp^2} + \frac{2}{p} \frac{d}{dp} \right) R(p) + \left(-\frac{l(l+1)}{p^2} + \frac{1}{p} - \frac{1}{4} \right) R(p) = 0$$

$$G'' - G' + \frac{1}{4} G + \frac{2}{p} \left(G' - \frac{1}{2} G \right) + \left(-\frac{l(l+1)}{p^2} + \frac{1}{p} - \frac{1}{4} \right) G = 0$$

$$G'' + G' \left(\frac{2}{p} - 1 \right) + G \left(\frac{1}{p} - \frac{1}{p} - \frac{l(l+1)}{p^2} + \frac{1}{p} - \frac{1}{4} \right) = 0$$

$$G'' + G' \left(\frac{2}{p} - 1 \right) + G \left(\frac{1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$G = p^l H \quad G' = l p^{l-1} H + p^l H' \quad G'' = l(l-1) p^{l-2} H + 2l p^{l-1} H' + p^l H''$$

$$G = H \quad G' = H' + \frac{l}{p} H \quad G'' = H'' + \frac{2l}{p} H' + \frac{l(l-1)}{p^2} H$$

$$H'' + \frac{2l}{p} H' + \frac{l(l-1)}{p^2} H + \left(\frac{2}{p} - 1 \right) \left(H' + \frac{l}{p} H \right) + H \left(\frac{1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$H'' + H' \left(\frac{2l}{p} + \frac{2}{p} - 1 \right) + H \left(\frac{2l}{p^2} + \frac{2l}{p^2} - \frac{l}{p} + \frac{1}{p} - \frac{l(l+1)}{p^2} \right) = 0$$

$$H'' + H' \left(\frac{2(l+1)}{p} - 1 \right) + H \left(\frac{1}{p} - \frac{(1+l)}{p} \right) = 0$$

$$H = \sum_n a_n p^n \Rightarrow H' = \sum_n a_n n p^{n-1} \quad H'' = \sum_n a_n n(n-1) p^{n-2}$$

$$\sum_n a_n n(n-1) p^{n-2} + \sum_n a_n n p^{n-1} \left(\frac{2(l+1)}{p} - 1 \right) + \sum_n a_n p^n \left(\frac{1}{p} - \frac{(1+l)}{p} \right) = 0$$

$$\sum_{n+1} a_{n+1} (n+1) n p^{n-1} + \sum_{n+1} a_{n+1} (n+1) p^{n-1} (2(l+1)) + \sum_n a_n p^{n-1} (1 - (1+l)) = 0$$

$$\sum_{n+1} a_{n+1} (n+1) (2(l+1) + n) = \sum_n a_n (1 + l + n - n) = 0$$

$$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{n+1+l-n}{(n+1)(2(l+1)+n)} = \frac{n-(n-1-l)}{(n+1)(2(l+1)+n)} \quad \text{Bas kuantum sayısı}$$

$n = n_r + l + 1$ tan sayı olmak
n bas kuantum sayısı

$$\frac{1}{2} \mu c^2 \frac{(Z\alpha)^2}{n^2} = E_n$$

Bölüm 13

maxwell 4 eq.

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{dB}{dt}$$

$$\nabla \times B = \frac{\mu_0 J}{c} + \frac{dE}{cdt}$$

$$E = -\nabla\phi + \frac{dA}{cdt}$$

$$B = \nabla \times A$$

$$\rightarrow \nabla \cdot E = \frac{\rho}{\epsilon_0} \Rightarrow \nabla \cdot (-\nabla\phi + \frac{dA}{cdt}) = \frac{\rho}{\epsilon_0} \Rightarrow (-\nabla^2\phi + \frac{d}{cdt}\nabla \cdot A) = \frac{\rho}{\epsilon_0} \Rightarrow -\nabla^2\phi = \frac{\rho}{\epsilon_0} \text{ (Poisson Denk)}$$

$$\nabla \cdot A = 0 \Rightarrow \text{Coulomb Ayarı}$$

$$\nabla \times B = \frac{\mu_0 J}{c} + \frac{dE}{cdt} \Rightarrow \nabla \times (\nabla \times \frac{dA}{cdt}) - \frac{dE}{cdt} = \frac{\mu_0 J}{c}$$

$$\partial_x (b \times c) = b \partial_c - c \partial_b$$

$$\nabla (\nabla A) - \nabla^2 A - \frac{dE}{cdt} = \frac{\mu_0 J}{c} \Rightarrow E = -\nabla\phi + \frac{dA}{cdt}$$

$$-\nabla (\nabla A) + \nabla^2 A - \frac{d}{cdt}(-\nabla\phi + \frac{dA}{cdt}) = \frac{\mu_0 J}{c} \Rightarrow \nabla (\nabla A - \frac{d\phi}{cdt}) + \nabla^2 A - \frac{d^2 A}{c^2 dt^2} = \frac{\mu_0 J}{c}$$

$$0 \Rightarrow \nabla A = \frac{d\phi}{cdt} \Rightarrow \text{Lorentz Ayarı}$$

$$\nabla \cdot E = \frac{\rho}{\epsilon_0} \Rightarrow \nabla \cdot (-\nabla\phi + \frac{dA}{cdt}) = \frac{\rho}{\epsilon_0}$$

$$-\nabla^2\phi + \frac{d}{cdt}\nabla \cdot A = \frac{\rho}{\epsilon_0} \Rightarrow -\nabla^2\phi + \frac{d^2\phi}{c^2 dt^2} = \frac{\rho}{\epsilon_0}$$

$$\nabla^2 A - \frac{d^2 A}{c^2 dt^2} = \frac{\mu_0 J}{c} \Rightarrow \nabla^2 A - \frac{d^2 A}{c^2 dt^2} = \frac{\mu_0 J}{c} \Rightarrow \square(A^\mu) = (c\rho, J^\mu)$$

De Lambert Operatörü

$$H\psi = E\psi \Rightarrow H = \frac{p^2}{2m} + V \Rightarrow H = \frac{1}{2\mu} (p + \frac{e}{c}A)^2 + e\phi$$

$$p = \frac{\hbar}{i} \frac{d}{dx} \Rightarrow \frac{\hbar}{i} \frac{d}{dr} = \frac{\hbar}{i} \vec{\nabla}$$

$$\frac{1}{2\mu} (\frac{\hbar}{i} \nabla + \frac{e}{c}A) (\frac{\hbar}{i} \nabla + \frac{e}{c}A) \psi = E\psi$$

$$\nabla \cdot A = 0 \text{ (Coulomb Ayarı)}$$

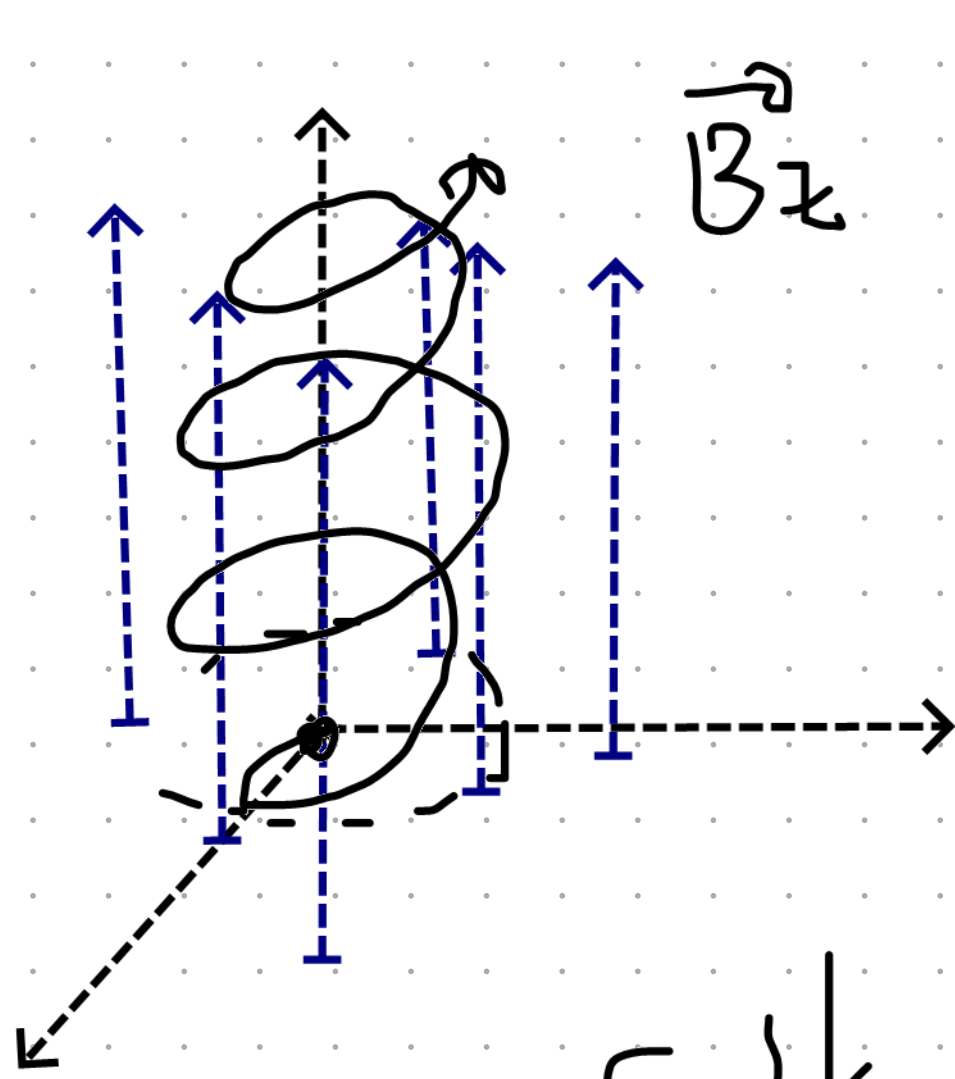
$$\Rightarrow \frac{1}{2\mu} (\frac{\hbar}{i} \nabla + \frac{e}{c}A) (\frac{\hbar}{i} \nabla \psi + \frac{e}{c}A\psi) = -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar}{i} \frac{e}{c} (\nabla A) \psi + \frac{\hbar}{i} \frac{e}{c} A (\nabla \psi) + \frac{e^2}{c^2} A^2 \psi = E\psi$$

$$\frac{1}{2\mu} (-\hbar^2 \nabla^2 \psi + 2 \frac{\hbar}{i} \frac{e}{c} A (\nabla \psi) + \frac{e^2}{c^2} A^2 \psi) = E\psi$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{\hbar}{i\mu c} (-\frac{1}{2}) (r \times B) (\nabla \psi) + \frac{e^2}{8\mu c^2} B^2 (x^2 + y^2) \psi$$

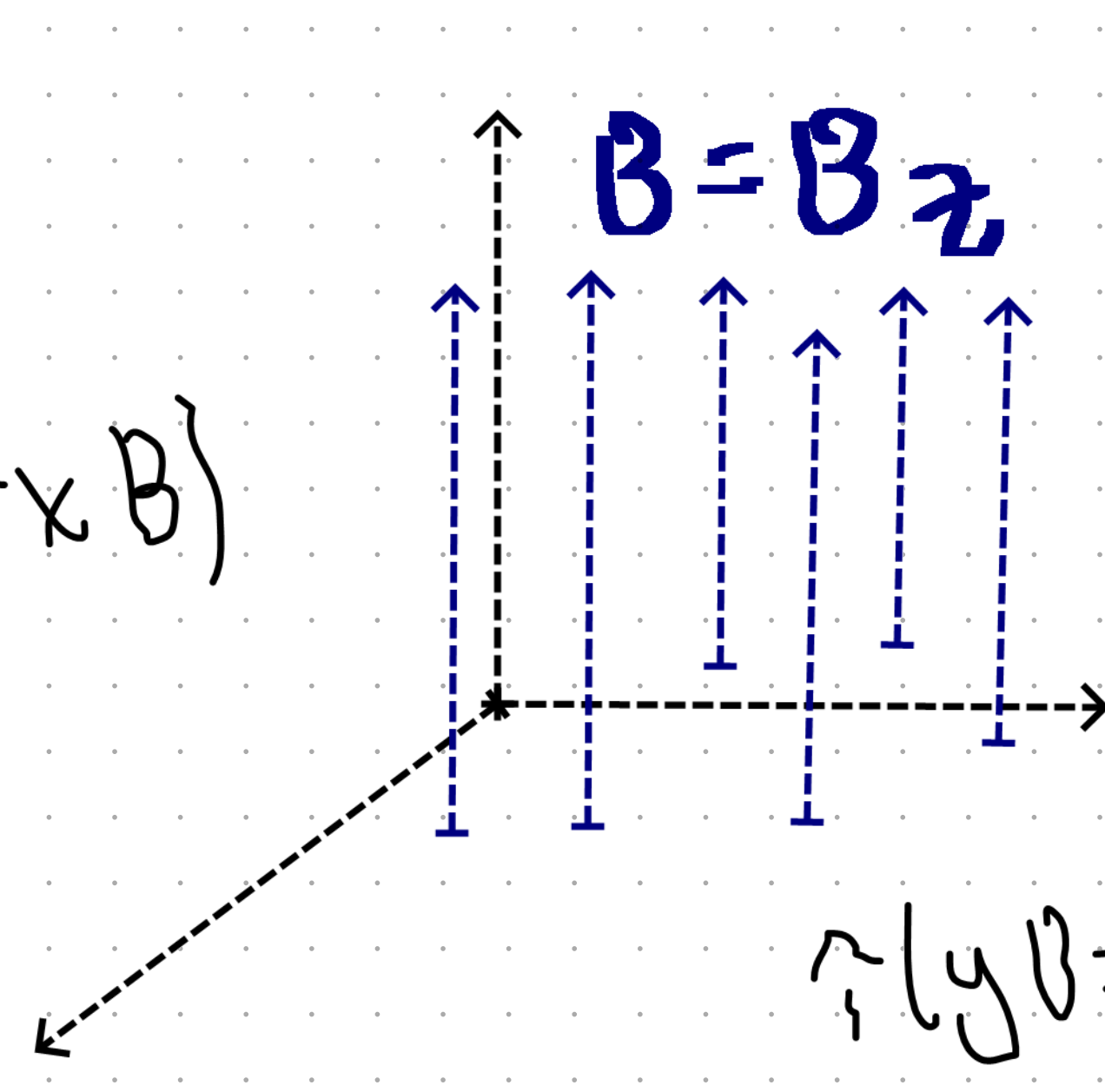
$$+ \frac{e}{2\mu c} (r \times \frac{\hbar}{i} \nabla) B \psi$$

$$= -\frac{\hbar^2}{2\mu} \nabla^2 \psi + \frac{eL}{2\mu c} B \psi + \frac{e^2}{8\mu c^2} B^2 (x^2 + y^2) \psi = E\psi$$



5 dk mola

$$A = -\frac{1}{2} (r \times B)$$



$$r \times B = \begin{vmatrix} i & j & k \\ x & y & z \\ B_x & B_y & B_z \end{vmatrix}$$

$$r \times B = \hat{i}(yB_z) - \hat{j}(xB_z) + \hat{k}(0)$$

$$\Rightarrow B_z (-y, x, 0)$$

$$-\frac{1}{2} (r \times B) = A \Rightarrow \frac{B_z}{2} (-y, x, 0) = \text{Coulomb Ayarı}$$

$$(-y, 0, 0), (0, x, 0) \text{ Landau Ayarı}$$

$$A = \frac{B_z}{2} (-y, x, 0) \quad A^2 = \frac{B_z^2}{4} (y^2 + x^2)$$

Bölüm 14

$$H\psi = E\psi \quad L^2 L_x L_y L_z \quad L_+ L_-$$

unverimlerin şeklinde yapalım.

$$H|u_n\rangle = E_n|u_n\rangle$$

$$\langle u_m | H | u_n \rangle = \langle u_m | E_n | u_n \rangle = E_n \langle u_m | u_n \rangle = E_n \delta_{mn} = (n + \frac{1}{2}) \hbar \omega \delta_{mn}$$

$$m_j \rightarrow n \rightarrow \begin{matrix} 0 & 1 & 2 & 3 & 4 \\ \hline 0 & \frac{1}{2} & 1 & \frac{3}{2} & 2 \\ 1 & \frac{3}{2} & 2 & \frac{5}{2} & 3 \\ 2 & \frac{5}{2} & 3 & \frac{7}{2} & 4 \\ 3 & \frac{7}{2} & 4 & \frac{9}{2} & 5 \\ 4 & \frac{9}{2} & 5 & \frac{11}{2} & 6 \end{matrix}$$

$$L^2 |l, m\rangle = \hbar^2 l(l+1) |l, m\rangle \quad |l, m\rangle = |u_n\rangle$$

$$L_z |l, m\rangle = \hbar m |l, m\rangle \Rightarrow u_n = ? \Rightarrow |u_n\rangle = |l', m'\rangle$$

$$\langle u_m | u_n \rangle = \langle l', m' | l, m \rangle$$

$$\Rightarrow \langle u_m | [L^2, L_z] | u_n \rangle$$

$$\Rightarrow \langle u_m | L^2 L_z - L_z L^2 | u_n \rangle$$

$$\langle l', m' | L^2 L_z - L_z L^2 | l, m \rangle = \hbar^2 \{ \underbrace{l'(l'+1)}_0 - \underbrace{l(l+1)}_0 \} \underbrace{\langle m' | L_z | m \rangle}_0 = 0$$

$$l' = l$$

$$\langle u_m | [L^2, L_z] | u_n \rangle = \langle l', m' | [L^2, L_z] | l, m \rangle$$

$$L_z |u_n\rangle = L_z |l, m\rangle = \hbar m |l, m\rangle$$

$$\langle u_m | L_z | u_n \rangle = \langle m' | L_z | m \rangle = \hbar m \langle m' | m \rangle \Rightarrow L_z = \hbar m \Rightarrow m_j = m$$

$$m = (2l+1) \quad l=1 \Rightarrow m = -1 \ 0 \ 1$$

$$m \rightarrow \begin{matrix} -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ 1 & 0 & 1 \end{matrix}$$

$$L_+ = L_x + iL_y$$

$$L_- = L_x - iL_y \Rightarrow \frac{L_+ + L_-}{2} = L_x \quad iL_y = \frac{L_+ - L_-}{2}$$

$$L_+ |l, m\rangle = \hbar \sqrt{l(l+1) - m(m+1)} |l, m+1\rangle \quad l=1 \Rightarrow m = (2l+1)$$

$$L_- |l, m\rangle = \hbar \sqrt{l(l+1) - m(m-1)} |l, m-1\rangle$$

$$\langle l, m' | L_+ | l, m \rangle = \sqrt{l(l+1) - m(m+1)} \hbar \langle m' | m+1 \rangle$$

$$l=1 \Rightarrow m = (-1, 0, 1)$$

$$\delta_{m, m+1}$$

$$\Rightarrow L_+ =$$

$$m \rightarrow \begin{matrix} -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ 1 & 0 & 1 \end{matrix}$$

$$\sqrt{2} \hbar \delta_{m, m-1}$$

$$L_- = \begin{matrix} m \rightarrow & -1 & 0 & 1 \\ m' \rightarrow & -1 & 0 & 1 \\ \hline & -1 & 0 & 1 \\ & 0 & \sqrt{2} & 0 \\ & 0 & 0 & \sqrt{2} \\ & +1 & 0 & 0 \end{matrix}$$

$$m = -1 \Rightarrow \hbar \sqrt{2-0} \langle m' | 0 \rangle \quad m = 1 \Rightarrow 0$$

$$m = 0 \Rightarrow \hbar \sqrt{2} \langle m' | 1 \rangle$$

$$L_+ = \begin{matrix} m \rightarrow & -1 & 0 & +1 \\ \downarrow m' & -1 & 0 & 0 \\ & 0 & \sqrt{2} & 0 \\ & 0 & 0 & \sqrt{2} \\ & +1 & 0 & 0 \end{matrix} \quad L_- = \begin{matrix} m \rightarrow & -1 & 0 & +1 \\ \downarrow m' & -1 & 0 & 0 \\ & 0 & \sqrt{2} & 0 \\ & 0 & 0 & \sqrt{2} \\ & +1 & 0 & 0 \end{matrix}$$

$$\frac{L_+ + L_-}{2} = L_x =$$

$$\begin{matrix} & -1 & 0 & +1 \\ \downarrow & -1 & \sqrt{2} & 0 \\ & 0 & \sqrt{2} & \sqrt{2} \\ & +1 & 0 & \sqrt{2} \end{matrix} = L_x$$

$$L_y = \frac{L_x - L_z}{2}$$

$$L^2 |l, m\rangle = L^2 |l, m\rangle = l(l+1)\hbar^2 |l, m\rangle$$

$$\langle l, m | L^2 | l, m \rangle = \langle l, m' | L^2 | l, m \rangle = l(l+1)\hbar^2 \langle l, m' | l, m \rangle$$

$$\Rightarrow l(l+1)\hbar^2 \langle l, m' | l, m \rangle = l(l+1)\hbar^2 \delta_{m'm} \Rightarrow$$

$$0 \Rightarrow 1 \Rightarrow m = -1$$

$$2\hbar^2 \begin{matrix} & -1 & 0 & +1 \\ \downarrow & -1 & 1 & 0 \\ & 0 & 1 & 1 \\ & +1 & 0 & 1 \end{matrix}$$

Harmonik operatörlerde azaltıcı ve artırıcı matrisler

$$A^+ |n\rangle = \sqrt{n+1} |n+1\rangle \quad A |n\rangle = \sqrt{n} |n-1\rangle$$

$$\langle m | A^+ | n \rangle = \sqrt{n+1} \langle m | n+1 \rangle$$

$$\langle m | A | n \rangle = \sqrt{n} \langle m | n-1 \rangle \delta_{m, n-1}$$

$$\sqrt{n+1} \delta_{m, n+1} = \begin{matrix} m \rightarrow & 0 & 1 & 2 & 3 & 4 \\ \downarrow n & 0 & \sqrt{1} & \sqrt{2} & \sqrt{3} & \sqrt{4} \end{matrix}$$

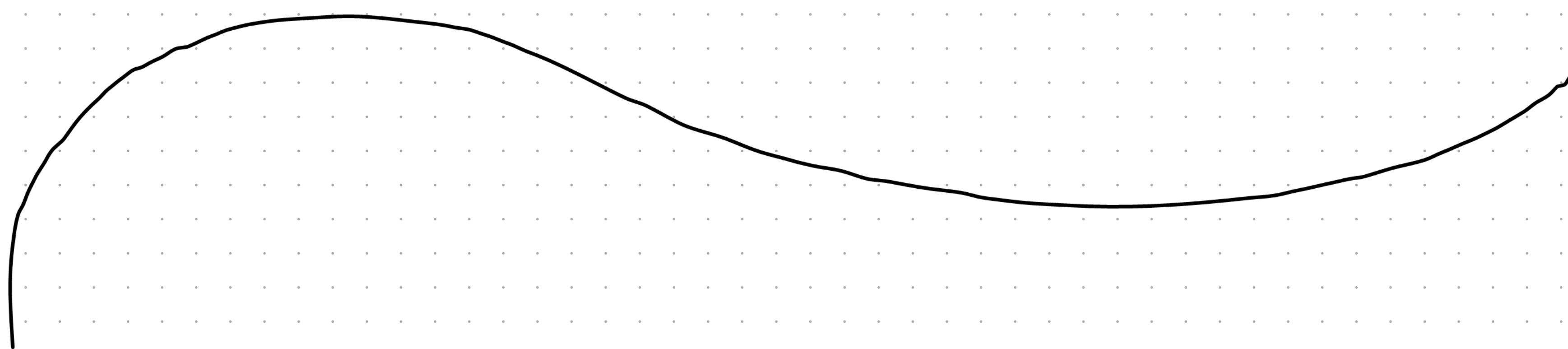
$$\sqrt{n} \delta_{m, n-1} = \begin{matrix} & 0 & 1 & 2 & 3 & 4 \\ \downarrow & 0 & \sqrt{1} & \sqrt{2} & \sqrt{3} & \sqrt{4} \end{matrix}$$

harmonik salınımlarda x

$$\hat{A}^+ = \sqrt{\frac{m\omega}{2\hbar}} x - \frac{ip}{\sqrt{2m\hbar\omega}}$$

$$\hat{A} = \sqrt{\frac{m\omega}{2\hbar}} x + \frac{ip}{\sqrt{2m\hbar\omega}}$$

$$\hat{A}^+ + \hat{A} = 2\sqrt{\frac{m\omega}{2\hbar}} x \Rightarrow \sqrt{\frac{2m\omega}{\hbar}} x = \hat{A}^+ + \hat{A} \Rightarrow x = \sqrt{\frac{\hbar}{2m\omega}} (\hat{A}^+ + \hat{A})$$



13. April 16

$$H|\psi\rangle = E|\psi\rangle \Rightarrow H = H_0 + \lambda H_1 + \lambda^2 H_2 \quad \psi = |\phi_n\rangle + \sum_{n \neq k} \lambda c_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^{(2)} |\phi_k\rangle$$

$$E = E_n^0 + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}$$

$$(H_0 + \lambda H_1) \left(|\phi_n\rangle + \sum_{n \neq k} \lambda c_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^{(2)} |\phi_k\rangle \right) = (E_n^0 + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}) \left(|\phi_n\rangle + \sum_{n \neq k} \lambda c_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^{(2)} |\phi_k\rangle \right)$$

$$\underbrace{H_0 |\phi_n\rangle}_{= E_n^0 |\phi_n\rangle} + \lambda \sum_{n \neq k} c_{nk}^{(1)} \underbrace{H_0 |\phi_k\rangle}_{= E_k^0 |\phi_k\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^{(2)} \underbrace{H_0 |\phi_k\rangle}_{= E_k^0 |\phi_k\rangle} + \lambda \underbrace{H_1 |\phi_n\rangle}_{= E_n^{(1)} |\phi_n\rangle} + \lambda \sum_{n \neq k} c_{nk}^{(1)} \underbrace{H_1 |\phi_k\rangle}_{= E_k^{(1)} |\phi_k\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^{(2)} \underbrace{H_1 |\phi_k\rangle}_{= E_k^{(1)} |\phi_k\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^{(2)} \underbrace{H_2 |\phi_k\rangle}_{= E_k^{(2)} |\phi_k\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^{(2)} \underbrace{H_2 |\phi_k\rangle}_{= E_k^{(2)} |\phi_k\rangle}$$

$$\lambda^0: \text{in } H_0 |\phi_n\rangle = E_n^0 |\phi_n\rangle \quad \lambda^1: \text{in } \sum_{n \neq k} c_{nk}^{(1)} H_0 |\phi_k\rangle + H_1 |\phi_n\rangle = \sum_{n \neq k} c_{nk}^{(1)} E_k^0 |\phi_k\rangle + E_n^{(1)} |\phi_n\rangle$$

$$\lambda^2: \text{in } \sum_{n \neq k} c_{nk}^{(2)} H_0 |\phi_k\rangle + \sum_{n \neq k} c_{nk}^{(1)} H_1 |\phi_k\rangle = \sum_{n \neq k} c_{nk}^{(2)} E_k^0 |\phi_k\rangle + \sum_{n \neq k} c_{nk}^{(1)} E_k^{(1)} |\phi_k\rangle + E_n^{(2)} |\phi_n\rangle$$

$$\sum_{n \neq k} c_{nk}^{(1)} H_0 |\phi_k\rangle + H_1 |\phi_n\rangle = \sum_{n \neq k} c_{nk}^{(1)} E_k^0 |\phi_k\rangle + E_n^{(1)} |\phi_n\rangle$$

$\langle \phi_n |$ ile genişlet

$$\sum_{n \neq k} c_{nk}^{(1)} \langle \phi_n | H_0 | \phi_k \rangle + \langle \phi_n | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk}^{(1)} E_k^0 \langle \phi_n | \phi_k \rangle + E_n^{(1)} \langle \phi_n | \phi_n \rangle$$

$$\langle \phi_n | H_1 | \phi_n \rangle = E_n^{(1)}$$

$\langle \phi_k |$ ile genişlet

$$\sum_{n \neq k} c_{nk}^{(1)} \langle \phi_k | H_0 | \phi_k \rangle + \langle \phi_k | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk}^{(1)} E_n^0 \langle \phi_k | \phi_k \rangle + E_n^{(1)} \langle \phi_k | \phi_n \rangle$$

$$\sum_{n \neq k} c_{nk}^{(1)} E_k^0 \langle \phi_k | \phi_k \rangle + \langle \phi_k | H_1 | \phi_n \rangle = \sum_{n \neq k} c_{nk}^{(1)} E_n^0$$

$$\frac{\langle \phi_k | H_1 | \phi_n \rangle}{E_n^0 - E_k^0} = \sum_{n \neq k} c_{nk}^{(1)}$$

$$\sum_{n \neq k} c_{nk}^{(2)} H_0 |\phi_k\rangle + \sum_{n \neq k} c_{nk}^{(1)} H_1 |\phi_k\rangle = \sum_{n \neq k} c_{nk}^{(2)} E_n^0 |\phi_k\rangle + \sum_{n \neq k} c_{nk}^{(1)} E_n^{(1)} |\phi_k\rangle + E_n^{(2)} |\phi_n\rangle$$

$\langle \phi_n |$ ile genişlet

$$\sum_{n \neq k} c_{nk}^{(2)} \langle \phi_n | H_0 | \phi_k \rangle + \sum_{n \neq k} c_{nk}^{(1)} \langle \phi_n | H_1 | \phi_k \rangle = \sum_{n \neq k} c_{nk}^{(2)} E_n^0 \langle \phi_n | \phi_k \rangle + \sum_{n \neq k} c_{nk}^{(1)} E_n^{(1)} \langle \phi_n | \phi_k \rangle + E_n^{(2)} \langle \phi_n | \phi_n \rangle$$

$$\sum_{n \neq k} c_{nk}^{(1)} \langle \phi_n | H_1 | \phi_k \rangle = E_n^{(2)}$$

$$\frac{\langle \phi_n | H_1 | \phi_k \rangle \langle \phi_n | H_1 | \phi_k \rangle}{E_n^0 - E_k^0} = E_n^{(2)}$$

$$\Rightarrow \frac{|\langle \phi_n | H_1 | \phi_k \rangle|^2}{E_n^0 - E_k^0} = E_n^{(2)}$$

Zaman Bağımlı Pert. Bölüm 21

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2\mu} \frac{\partial^2 \Psi}{\partial x^2} = E\Psi$$

$$i\hbar \frac{\partial \Psi}{\partial t} = E\Psi \Rightarrow \frac{\partial \Psi}{\partial t} = \frac{E}{i\hbar} \Psi \Rightarrow \frac{\partial \Psi}{\Psi} = \frac{E}{i\hbar} dt \Rightarrow \ln \Psi = \frac{E}{i\hbar} t + \ln C$$

$$\Rightarrow \ln \frac{\Psi}{C} = \frac{E}{i\hbar} t \Rightarrow \ln \Psi = \ln \left(\frac{E}{i\hbar} t \right)$$

$$e^{\ln \Psi} = e^{\ln C} e^{Et/i\hbar}$$

$$H\Psi = E\Psi$$

$$\Psi = C e^{Et/i\hbar}$$

$$\Psi_n = C_n e^{Et/i\hbar} \Rightarrow |\Phi_n\rangle = \frac{1}{C_n} e^{Et/i\hbar}$$

$$i\hbar \left(\frac{\partial}{\partial t} \sum_n C_n e^{-iEt/\hbar} \right) = -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial x^2} \sum_n C_n e^{-iEt/\hbar}$$

$$i\hbar \left(\sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} - \frac{iE_n^0}{\hbar} \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \right) = -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial x^2} \sum_n C_n(t) e^{-iE_n^0 t/\hbar}$$

$$i\hbar \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} + E_n^0 \sum_n C_n(t) e^{-iE_n^0 t/\hbar} = E_n^0 \sum_n C_n(t) e^{-iE_n^0 t/\hbar} + (N|V|0) \text{ katkısı olmalı}$$

$$H\Psi = E\Psi \Rightarrow H = H_0 + N|H| \Rightarrow H = H_0 + N|V|0$$

$$i\hbar \frac{\partial \Psi}{\partial t} = (H_0 + N|V|) \Psi \Rightarrow i\hbar \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} = N \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \text{ vol } |\Phi_n\rangle$$

$\Rightarrow \langle \Phi_m |$ ile çarpalım

$$i\hbar \sum_n \dot{C}_n(t) e^{-iE_n^0 t/\hbar} \langle \Phi_m | \Phi_n \rangle = N \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$\sum_n \dot{C}_m(t) e^{-iE_m^0 t/\hbar} = \frac{N}{i\hbar} \sum_n C_n(t) e^{-iE_n^0 t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$\sum_n \dot{C}_m(t) = \frac{N}{i\hbar} \sum_n C_n(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$\dot{C}_m(t) = N^0 \dot{C}_m^0 + N^1 \dot{C}_m^1(t) + N^2 \dot{C}_m^2 \dots$$

$$C_n(t) = N^0 C_n^0(t) + N^1 C_n^1(t) + N^2 C_n^2(t)$$

$$\dot{C}_m^0(t) = \frac{N}{i\hbar} C_n^0(t) e^{-iE_n^0 t/\hbar} = \delta_{mn}$$

$$C_m^1(t) = \frac{N}{i\hbar} C_n^0(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$C_m^2(t) = \frac{N}{i\hbar} C_n^1(t) e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$\frac{d}{dt} C_m^1(t) = \frac{N}{i\hbar} e^{-i(E_n^0 - E_m^0)t/\hbar} \langle \Phi_m | V | \Phi_n \rangle$$

$$C_m^1(t) = \frac{N}{i\hbar} \int e^{-i(E_n^0 - E_m^0)t/\hbar} dt \langle \Phi_m | V | \Phi_n \rangle$$

Dejenerat Perturbasyon Teorisi

$$H\Psi = E\Psi \quad H = H_0 + \lambda H_1 + \lambda^2 H_2$$

$$|\Psi\rangle = |\phi_n\rangle + \left(\sum_{n \neq k} \lambda C_{nk}^{(1)} + \sum_{n \neq k} \lambda^2 C_{nk}^{(2)} \right) |\phi_k\rangle$$

$$|\phi_n\rangle + \sum_{n \neq k} \lambda C_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 C_{nk}^{(2)} |\phi_k\rangle$$

$$H = H_0 + \lambda H_1 + \lambda^2 H_2 \quad E = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}$$

$$(H_0 + \lambda H_1) (|\phi_n\rangle + \sum_{n \neq k} \lambda C_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 C_{nk}^{(2)} |\phi_k\rangle) = (E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}) (|\phi_n\rangle + \sum_{n \neq k} \lambda C_{nk}^{(1)} |\phi_k\rangle + \sum_{n \neq k} \lambda^2 C_{nk}^{(2)} |\phi_k\rangle)$$

$$H_0 |\phi_n\rangle + \sum_{n \neq k} \lambda C_{nk}^{(1)} H_0 |\phi_k\rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(2)} H_0 |\phi_k\rangle + \lambda H_1 |\phi_n\rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(1)} H_1 |\phi_k\rangle = E_n^{(0)} |\phi_n\rangle + \lambda \sum_{n \neq k} C_{nk}^{(1)} E_n^{(0)} |\phi_k\rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(2)} E_n^{(0)} |\phi_k\rangle + \lambda E_n^{(1)} |\phi_n\rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(1)} E_n^{(1)} |\phi_k\rangle + \lambda^2 E_n^{(2)} |\phi_n\rangle$$

$\langle \phi_n |$ ile çarpalım

$$\langle \phi_n | H_0 | \phi_n \rangle + \sum_{n \neq k} \lambda C_{nk}^{(1)} \langle \phi_n | H_0 | \phi_k \rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(2)} \langle \phi_n | H_0 | \phi_k \rangle + \lambda \langle \phi_n | H_1 | \phi_n \rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(1)} \langle \phi_n | H_1 | \phi_k \rangle =$$

$$= E_n^{(0)} \langle \phi_n | \phi_n \rangle + \lambda \sum_{n \neq k} C_{nk}^{(1)} E_n^{(0)} \langle \phi_n | \phi_k \rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(2)} E_n^{(0)} \langle \phi_n | \phi_k \rangle + \lambda E_n^{(1)} \langle \phi_n | \phi_n \rangle + \lambda^2 \sum_{n \neq k} C_{nk}^{(1)} E_n^{(1)} \langle \phi_n | \phi_k \rangle + \lambda^2 E_n^{(2)} \langle \phi_n | \phi_n \rangle$$

$$n^0 \Rightarrow \langle \phi_n | H_0 | \phi_n \rangle = E_n^{(0)} \langle \phi_n | \phi_n \rangle$$

$$n^1 \Rightarrow \sum_{n \neq k} \lambda C_{nk}^{(1)} \langle \phi_n | H_0 | \phi_k \rangle + \langle \phi_n | H_1 | \phi_n \rangle = \sum_{n \neq k} \lambda C_{nk}^{(1)} E_n^{(0)} \langle \phi_n | \phi_k \rangle + E_n^{(1)} \langle \phi_n | \phi_n \rangle$$

$$\langle \phi_n | H_1 | \phi_n \rangle = E_n^{(1)} \langle \phi_n | \phi_n \rangle$$

$$\sum_j \alpha_j h_{ji} = E_n^{(1)} \alpha_i \quad j, i = 1, 2$$

$$\sum_{j=1}^2 \alpha_j h_{ji} = E_n^{(1)} \alpha_i \Rightarrow \alpha_j h_{1j} = E_n^{(1)} \alpha_j \quad j=1, 2$$

$$\alpha_j h_{2j} = E_n^{(1)} \alpha_j$$

$$\Rightarrow \alpha_1 h_{11} + \alpha_2 h_{12} = E_n^{(1)} \alpha_1 + E_n^{(1)} \alpha_2$$

$$\alpha_1 h_{21} + \alpha_2 h_{22} = E_n^{(1)} \alpha_1 + E_n^{(1)} \alpha_2$$

$$\begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

Degenerierte Pert.

$$H\Psi = E\Psi \Rightarrow H = H_0 + \lambda H_1 + \lambda^2 H_2 \quad \Psi = |\phi_n\rangle + \lambda \sum_{n \neq k} c_{nk}^1 |\phi_k\rangle + \lambda^2 \sum_{n \neq k} c_{nk}^2 |\phi_k\rangle$$

$$E = E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2$$

$$\text{Degenerierte } |\phi_n^i\rangle + \sum_{n \neq k} \lambda c_{nk}^1 |\phi_k^i\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^2 |\phi_k^i\rangle$$

$$(H_0 + \lambda H_1) (|\phi_n^i\rangle + \sum_{n \neq k} \lambda c_{nk}^1 |\phi_k^i\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^2 |\phi_k^i\rangle) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2) (|\phi_n^i\rangle + \sum_{n \neq k} \lambda c_{nk}^1 |\phi_k^i\rangle + \sum_{n \neq k} \lambda^2 c_{nk}^2 |\phi_k^i\rangle)$$

$$\underline{H_0 |\phi_n^i\rangle} + \lambda \sum_{n \neq k} c_{nk}^1 \underline{H_0 |\phi_k^i\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^2 H_0 |\phi_k^i\rangle + \lambda \underline{H_1 |\phi_n^i\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^1 H_1 |\phi_k^i\rangle =$$

$$\underline{E_n^0 |\phi_n^i\rangle} + \sum_{n \neq k} \lambda c_{nk}^1 \underline{E_n^0 |\phi_k^i\rangle} + \sum_{n \neq k} \lambda^2 c_{nk}^2 E_n^0 |\phi_k^i\rangle + \lambda \underline{E_n^1 |\phi_n^i\rangle} + \lambda^2 \sum_{n \neq k} c_{nk}^1 E_n^1 |\phi_k^i\rangle + \lambda^2 E_n^2 |\phi_n^i\rangle$$

$$\lambda^0 \Rightarrow H_0 |\phi_n^i\rangle = E_n^0 |\phi_n^i\rangle$$

$$\lambda^1 \Rightarrow \sum_{n \neq k} c_{nk}^1 H_0 |\phi_k^i\rangle + H_1 |\phi_n^i\rangle = \sum_{n \neq k} c_{nk}^1 E_n^0 |\phi_k^i\rangle + E_n^1 |\phi_n^i\rangle$$

$$\langle \phi_n^j | \text{ile gewichtet} \sum_{n \neq k} c_{nk}^1 \langle \phi_n^j | H_0 |\phi_k^i\rangle + \langle \phi_n^j | H_1 |\phi_n^i\rangle = \sum_{n \neq k} c_{nk}^1 E_n^0 \langle \phi_n^j | \phi_k^i\rangle + E_n^1 \langle \phi_n^j | \phi_n^i\rangle$$

$$\langle \phi_n^j | H_1 |\phi_n^i\rangle = E_n^1 \langle \phi_n^j | \phi_n^i\rangle$$

$$|\phi_n^i\rangle = \sum_j \alpha_j |\phi_n^j\rangle$$

$$\langle \phi_n^j | = \sum_i \alpha_i^* \langle \phi_n^i |$$

$$\sum_j \alpha_j \langle \phi_n^j | H_1 |\phi_n^i\rangle = \sum_j \alpha_j E_n^1 \langle \phi_n^j | \phi_n^i\rangle$$

$$\sum_j \alpha_j h_{ji}^n = \alpha_i E_n^1$$

$$i, j = 1, 2 \quad \begin{matrix} 11 & 12 & 21 & 22 \end{matrix}$$

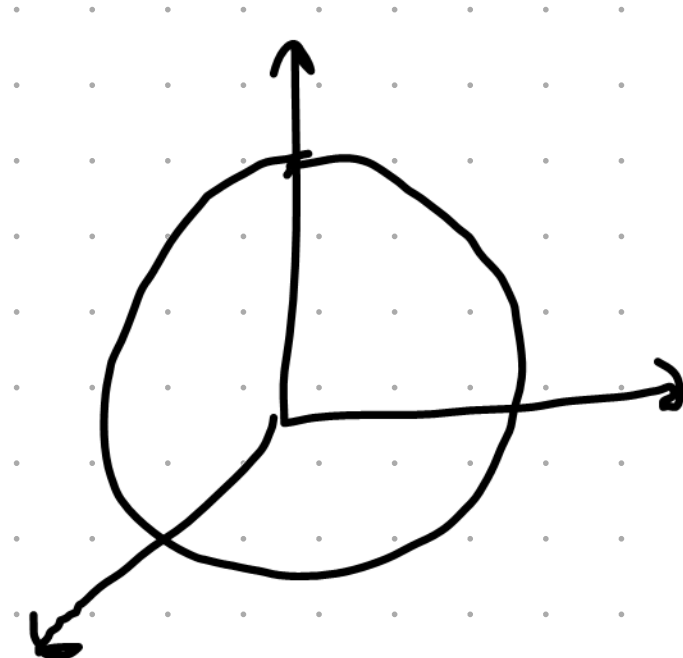
$$j=1 \Rightarrow \alpha_1 h_{11} + \alpha_2 h_{12} = \alpha_1 E_n^1 + \alpha_2 E_n^2$$

$$\alpha_1 h_{21} + \alpha_2 h_{22} = \alpha_1 E_n^1 + \alpha_2 E_n^2$$

$$\Rightarrow \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = E_n^1 \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

Stark Effekt

$$H_1 = e \vec{E} \cdot \vec{r}$$



$\vec{E} = E_z \hat{z}$ Homogen
Elektrisch Feld

$$\langle \phi_n^j | H_1 | \phi_n^i \rangle = h_{ji} = e E \langle \phi_n^j | z | \phi_n^i \rangle$$

$$e E \int d^3 r \phi_n^{j*} z \phi_n^i$$

$$e E \int_{-\infty}^{+\infty} d^3 r |\phi_n^i|^2 z = 0$$

cift fonk

$$E_n^2 = \frac{|\langle \phi_k^0 | H_1 | \phi_n^0 \rangle|^2}{E_k^0 - E_n^0}$$

$$E_n^2 = \frac{|\langle \phi_k^0 | H | \phi_n^0 \rangle|^2}{E_k^0 - E_n^0}$$

$$E_{n0}^2 = \frac{|\langle \phi_{nlm}^0 | H | \phi_{100}^0 \rangle|^2}{E_{nlm} - E_{100}} \quad \phi_{100} = R_{10} \psi_{00}$$

$$\hat{r} = r \cos \theta \hat{z}$$

$$R_{10}(r) \psi_{00}(\theta, \phi)$$

$$\psi_{00} = \frac{1}{\sqrt{4\pi}}$$

$$\int d^3r \phi_{nlm}^{0*}(r) (r \cos \theta) \phi_{100}^0(r)$$

$$\int d\Omega \int r^2 dr \Rightarrow \int d\Omega \int r^2 dr \phi_{nlm}^{0*}(r \cos \theta) \phi_{100}^0$$

$$\int d\Omega \int dr r^3 \underbrace{\phi_{nlm}^{0*}}_{R_{nl} \psi_{lm}} \cos \theta \underbrace{\phi_{100}^0}_{R_{10} \psi_{00}}$$

$$\Rightarrow \int d\Omega \psi_{lm}^*(\theta, \phi) \cos \theta \psi_{00}(\theta, \phi) \int dr r^3 R_{nl}^*(r) R_{10}(r)$$

$$\psi_{00} = \frac{1}{\sqrt{4\pi}} \quad \psi_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$\Rightarrow \int d\Omega \psi_{lm}^* \cos \theta \psi_{00}$$

$$\Rightarrow \int d\Omega \psi_{lm}^* \cos \theta \frac{\psi_{10}}{\sqrt{4\pi}} \sqrt{\frac{4\pi}{3}} \frac{1}{\cos \theta}$$

$$\Rightarrow \psi_{00} = \frac{\psi_{10}}{\sqrt{4\pi}} \sqrt{\frac{4\pi}{3}} \frac{1}{\cos \theta}$$

$$\frac{1}{\sqrt{4\pi}} \int d\Omega \psi_{lm}^* \psi_{10} \sqrt{\frac{4\pi}{3}}$$

$$\Rightarrow \frac{1}{\sqrt{3}} \int d\Omega \psi_{lm}^* \psi_{10} \quad (l=m=0) \quad \delta_{l1} \delta_{m0}$$

$$\frac{1}{\sqrt{3}} \int r^3 dr R_{nl} R_{10} = \frac{\delta_{l1} \delta_{m0}}{\sqrt{3}} \int r^3 dr R_{nl} R_{10}$$

$$R_{nl} r = L_{n-l-1}^{2l+1} (\rho) e^{-\rho/2} \quad \rho = \left(\frac{2\mu E}{\hbar^2} \right)^{1/2} r$$

